

UAV Resource Allocation System

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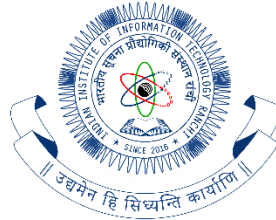
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UAV Resource Allocation System: Optimized Logistics for Efficient Delivery

Executive Summary

Efficient allocation of Unmanned Aerial Vehicles (UAVs) is crucial in scenarios where resources need to be transported from a central base to multiple outposts, each with varying demands and deadlines. This system, implemented in C++ using Object-Oriented Programming (OOP) principles, leverages graph-based algorithms like Dijkstra's for optimal route calculation, priority ordering for timely delivery scheduling, and capacity-based allocation to ensure UAVs are efficiently utilized while adhering to operational constraints such as weight, speed, and travel distance.

The goal is to minimize delivery times, meet deadlines, and ensure outposts receive their required resources, either fully or partially, using a minimal number of UAVs.

Table of Contents

1. Introduction
 2. Problem Definition
 3. System Design and Implementation
 4. Algorithmic Approach
 - o Graph Representation and Dijkstra's Algorithm
 - o Priority Ordering
 - o UAV Selection and Allocation
 - o Weight and Delivery Calculations
 5. Optimization and Results
 6. Code Architecture
 7. Future Enhancements
 8. Conclusion
-

1. Introduction

Unmanned Aerial Vehicles (UAVs) play an integral role in modern logistics by enabling quick and precise deliveries. However, managing the allocation of UAVs to meet multiple outposts' demands—while factoring in constraints like capacity, speed, range, and time limits—is a challenging task. This system, implemented in C++, uses object-oriented principles and graph-based algorithms to optimize deliveries, ensuring that resources are transported with minimal delays and maximum efficiency.

2 Motivation

In scenarios such as disaster relief, where rapid and accurate delivery of resources can save lives, the importance of efficient UAV allocation cannot be overstated. Similarly, in remote areas or conflict zones, UAVs can deliver essential supplies where traditional logistics methods fail. The motivation for this project stems from the need to:

1. Address real-world challenges of constrained resources.
2. Optimize logistics systems for timely deliveries.

3. Provide a modular and scalable approach adaptable to different scenarios.

3 Models and Problem Definition

The system is designed to manage UAVs and outposts under the following constraints:

3.1 UAV Characteristics

Each UAV is defined by:

- **Weight Capacity:** Maximum load a UAV can carry.
- **Speed:** Determines travel time.
- **Operational Range:** Maximum distance the UAV can travel for a round trip.

3.2 Outpost Requirements

Each outpost specifies:

- **Time Window:** The deadline by which resources must be delivered.
- **Total Weight:** The quantity of resources required.

3.3 Objectives

- Minimize total delivery time.
 - Efficiently allocate UAVs while adhering to their constraints.
 - Fully or partially fulfill outpost demands.
-

4. Literature Review

4.1 Existing Solutions:

- Graph-based algorithms such as Dijkstra's and A* have been widely used for routing optimization in logistics.
- Dynamic programming methods are effective for solving scheduling problems but often fail to scale well with real-time constraints.
- Resource Allocation for UAV-Enabled MEC Systems
- **UAV-Enabled MEC:** Utilizes UAVs for Mobile Edge Computing (MEC), enhancing computation by processing data closer to users.
- **Computation Efficiency Maximization:** Aims to optimize UAV placement, task offloading, and resource allocation to improve computational efficiency.
- **Mathematical Optimization:** Introduces a model that factors in UAV positions, task offloading decisions, and resource distribution among UAVs and ground users.
- Constraints: UAVs have limited capacity, and communication channels have latency. Tasks have latency and computational requirements.
- Solution Approach: Likely proposes heuristic or optimization techniques to balance multiple constraints and optimize resource allocation.
- **UAV Placement:** Optimal placement minimizes latency and enhances task offloading efficiency.
- **Task Offloading:** Decides which tasks should be offloaded based on UAV capabilities, communication costs, and latency requirements.
- **Dynamic Resource Allocation:** Adapts to changes in workload and network conditions to optimize resource usage.

4.2 Limitations of Current Approaches:

1. Lack of real-time adaptability.
2. Difficulty in handling multi-criteria optimization (e.g., time and weight).
3. Inadequate modularity for scaling.

This project integrates graph algorithms with object-oriented design to overcome these challenges.

5. System Design and Implementation

5.1 Object-Oriented Design

The solution is modular, employing an object-oriented approach in C++ with the following classes:

1. **UAV**
 - **Attributes:** id, capacity, speed, max_distance.
 - **Methods:** `calculate_time()`, `can_fulfill()`.
 2. **Outpost**
 - **Attributes:** id, deadline, weight_needed.
 - **Methods:** `priority_score()`.
 3. **Base**
 - **Attributes:** List of UAVs, adjacency matrix for distances.
 - **Methods:** `assign_uavs()`, `generate_priority_queue()`.
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5. Purpose and Methodology

Purpose:

The purpose of this project is to develop a robust system that uses UAVs to deliver resources efficiently under complex constraints.

Methodology:

1. **Graph Representation:** Model the base, outposts, and paths as nodes and edges in a graph.
2. **Shortest Path Calculation:** Use Dijkstra's Algorithm to find optimal routes for UAVs.
3. **Priority-Based Allocation:** Assign UAVs based on urgency and capacity.
4. **Modular Design:** Use object-oriented programming for scalability and maintainability.

7 Algorithmic Approach

7.1 Graph Representation and Dijkstra's Algorithm

The system is modeled as a weighted graph:

- **Nodes:** Represent the base and outposts.
- **Edges:** Represent paths between nodes, weighted by travel time.

7.2 Dijkstra's Algorithm is used to compute the shortest path between the base and outposts:

- **Initialization:** Distance array is set to infinity, with the base's distance as 0.
 - **Processing:** Nodes with the smallest distances are processed using a priority queue. For each node, distances to neighboring nodes are updated.
-

7.3 Priority Ordering of Outposts

Outposts are ranked by a priority score:

$$\text{Priority Score} = \frac{1}{\text{Time Remaining}}$$

$\text{Priority Score} = \frac{1}{\text{Time Remaining}}$

The priority queue ensures that the UAVs fulfill the most urgent demands first.

7.4 UAV Selection and Allocation

The allocation process follows these steps:

1. **Check Feasibility:** Evaluate UAVs based on their weight capacity, travel distance, and round-trip time.
 2. **Assign UAVs:** UAVs are selected based on the urgency of the outpost and their availability.
 3. **Partial vs. Complete Fulfillment:**
 - **Complete Fulfillment:** If a UAV's capacity meets the outpost's full demand, it's assigned entirely.
 - **Partial Fulfillment:** If a UAV's capacity is exceeded, the demand is split across multiple UAVs.
-

7.5 Weight and Delivery Calculations

- **Complete Fulfillment:** A UAV is assigned to an outpost if its capacity is sufficient for the full demand.
 - **Partial Fulfillment:** Multiple UAVs may be used if a single UAV's capacity cannot handle the full weight. The UAVs are prioritized based on their speed, ensuring faster deliveries.
-

8.Key Components

8.1. UAV Characteristics:

Each UAV has the following properties:

- Capacity (C): The maximum weight a UAV can carry.
- Speed (S): The speed at which the UAV travels.
- Maximum Distance (D_max): The maximum distance a UAV can travel in a single trip.
- Fuel Consumption (F): The amount of fuel consumed by the UAV, which is proportional to the distance traveled.
- Total Time (T): The total time elapsed since the UAV started its operation.
- Available Capacity (C_avail): The remaining weight the UAV can carry.

8.2. Outpost Characteristics:

Each outpost has:

- Required Weight (W): The amount of resources needed by the outpost.
- Deadline (D): The time by which the outpost needs the delivery to be made.
- Coordinates (x, y): The geographical location of the outpost.

8.3. Base Station Characteristics:

The base station is located at coordinates (x_0, y_0) and serves as the starting point for the UAVs.

9.Mathematical Formulation

9.1. Distance Calculation

The distance between two points (x_1, y_1) and (x_2, y_2) on the plane is calculated using the Euclidean distance formula:

$$\text{Distance}(x_1, y_1, x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This equation is used to determine the distance between:

- The UAV's current position and the outpost.
- The outpost and the base station (for the return journey).

9.2. Time Calculation

The time taken by a UAV to travel a given distance d at speed S is calculated as:

$$\text{Time}(d) = d / S$$

This is used to compute the time required for the UAV to travel to an outpost and back to the base.

9.3. Feasibility Condition

To determine if a UAV can fulfill a delivery, the following conditions must be satisfied:

- Capacity: The weight to be delivered must be less than or equal to the UAV's available capacity:
$$W \leq C_{\text{avail}}$$
- Distance: The UAV must be able to travel the required distance without exceeding its maximum range:
$$d \leq D_{\text{max}}$$
- Time Constraint: The UAV must be able to complete the delivery within the outpost's deadline:
$$\text{Time}(d) \leq D - T_{\text{current}}$$

where T_{current} is the UAV's total time used up until the current operation.

9.4. Outpost Priority Score

Each outpost is assigned a priority score based on its deadline, which is inversely proportional to the urgency of the delivery. The priority score for outpost i is given by:

$$P_i = 1 / D_i$$

Outposts with smaller deadlines (i.e., more urgent deliveries) will have higher priority.

9.5. Resource Allocation Process

UAVs are assigned to outposts based on a sorted list of outposts by their priority score. For each outpost, the following steps are performed:

- A search is conducted to find the UAV that can fulfill the outpost's delivery within the required time and weight constraints.
- The UAV with the minimum total time to complete the delivery (including return time) is selected.
- The UAV makes the delivery, updating its available capacity, total time, and fuel consumption.
- If the UAV can complete the task, the outpost's remaining required weight is reduced. If the outpost is fulfilled, the UAV returns to the base for refueling and reassignment.

9.6. Fuel Consumption

The UAV's fuel consumption is proportional to the distance traveled. For a UAV traveling a distance d , the fuel consumed is:

$$F = d$$

This accounts for both the trip to the outpost and the return trip.

9.7. UAV Time and Capacity Updates

After each delivery, the following updates are made:

- Total Time: The UAV's total time is updated as:
$$T_{\text{new}} = T_{\text{current}} + \text{Time}(d_{\text{to_outpost}}) + \text{Time}(d_{\text{return}})$$
- Available Capacity: The UAV's available capacity is updated as:
$$C_{\text{avail}} = C_{\text{avail}} - W_{\text{delivered}}$$

9.8. Optimization Objective

The objective of this model is to maximize the number of fulfilled outposts while minimizing the total fuel consumed and time used by the UAVs. This can be formalized as an optimization problem:

$$\max(\sum_{i=1}^n \text{Fulfilled}_i)$$
$$\text{subject to: } \min(\sum_{i=1}^n F_i + T_i)$$

where:

- Fulfilled_i is a binary variable indicating whether outpost i has been fulfilled (1 if fulfilled, 0 otherwise),
- F_i is the fuel consumed for the delivery to outpost i ,
- T_i is the total time taken for the delivery and return for outpost i .

10. Optimization and Results

10.1 Optimized Allocation

- **Path Optimization:** Dijkstra's algorithm ensures the UAVs use the most efficient paths, minimizing travel time.
- **Priority Fulfillment:** Outposts are processed based on their urgency, ensuring timely deliveries.

10.2 Performance Metrics

- **Total Delivery Time:** The total time taken by UAVs to fulfill the demands.
- **Percentage of Fulfilled Demands:** The proportion of the total demand successfully fulfilled.
- **UAV Utilization Efficiency:** The efficiency with which each UAV is used based on its capacity and availability.

11. Code Architecture

11.1 Key Functions

- `calculate_shortest_path()`: Implements Dijkstra's algorithm to find the shortest paths.
- `assign_uavs()`: Allocates UAVs based on priority and availability.
- `fulfill_demand()`: Handles both complete and partial deliveries.

11.2 Sample Code

```
cpp
Copy code
void Base::assign_uavs() {
    // Assign UAVs based on priority and availability
    for (auto &outpost : priority_queue) {
        // Check if the UAV can fulfill the demand
        UAV *selected_uav = select_uav(outpost);
        if (selected_uav != nullptr) {
            fulfill_demand(selected_uav, outpost);
        }
    }
}
```

12. Future Enhancements

1. **Dynamic Allocation:** Real-time updates for UAV availability or outpost requirements.
2. **Scalability:** Expand to handle multiple bases and hundreds of outposts.
3. **Advanced Algorithms:** Incorporate machine learning for demand forecasting and optimization.

13. Conclusion

This UAV resource allocation system efficiently solves the logistics challenge of delivering resources to multiple outposts while respecting UAV constraints. The use of Dijkstra's algorithm for path optimization, priority queues for scheduling, and capacity-based allocation ensures that the UAVs are used optimally. The system can be scaled and enhanced to accommodate dynamic environments, providing a robust foundation for UAV logistics management.

14 Behind-the-Scenes Process

Here's a detailed breakdown of how the system operates behind the scenes:

14.1. Initialization

- **Graph Creation:** A weighted graph is created where nodes represent the base and outposts, and edges represent travel times based on distances and UAV speeds.
- **UAV and Outpost Setup:** UAVs are initialized with their attributes (capacity, speed, range), and outposts are initialized with their resource demands and deadlines.

14.2. Priority Queue Generation

- **Outpost Scoring:**

A priority score is calculated for each outpost based on its remaining time to deadline:

Priority Score = $1/\text{Time Remaining}$

$$\text{Priority Score} = \frac{1}{\text{Time Remaining}}$$

- **Sorting:**

Outposts are sorted into a priority queue, ensuring UAVs attend to the most urgent needs first.

14.3. Path Calculation

- **Dijkstra's Algorithm:**
For each UAV, the shortest path to each outpost is computed.
 - The algorithm initializes all distances as infinity, with the base's distance set to 0.
 - Using a priority queue, nodes are processed to relax edges, ensuring optimal routes.
- **Travel Time Estimation:**
Travel time for each path is calculated based on UAV speed and distance.

14.4 Resource Allocation

- **Outpost Evaluation:**

Each outpost in the priority queue is assessed to determine if a UAV can fulfill its demand. Factors considered include:

- Remaining UAV capacity.
- Remaining outpost demand.
- UAV's remaining range and delivery deadlines.

- **Complete vs. Partial Fulfillment:**

- **Complete Fulfillment:** If a UAV's capacity can handle the outpost's demand, it is assigned directly.
- **Partial Fulfillment:** If demand exceeds a single UAV's capacity, it is split among multiple UAVs.

- **UAV Selection:**

UAVs are chosen to minimize time delays, prioritizing faster and more capable UAVs for urgent deliveries.

14.5. Delivery Execution

- **Order of Deliveries:**

Each UAV follows its assigned path, making deliveries sequentially. Travel and delivery times are logged for monitoring.

- **Post-Delivery Updates:**

After completing deliveries:

- UAV fuel levels and availability are updated.
- Outpost demand statuses are marked as fulfilled or pending.

14.6. Reassignment and Refueling

- **Return and Refuel:**

If all UAVs are occupied or need refueling:

- They return to the base, where their fuel is replenished.
- A time delay is logged for the refueling process.

- **Reassignment:**

After refueling, UAVs are reassigned based on the updated priority queue of pending outposts.

14.7. Logging and Monitoring

- **Status Updates:**

- The system continuously monitors the status of UAVs and outposts.
- Logs include delivery times, pending demands, UAV utilization, and fuel levels.

- **Performance Metrics:**

Metrics like total delivery time, fulfilled demands percentage, and UAV efficiency are calculated to assess system performance.

14.8. Iterative Process

- The system continues to operate in an iterative manner, adjusting to dynamic changes until all outposts have been served.

This behind-the-scenes process ensures that UAV resources are utilized optimally and deadlines are met efficiently.

Illustrative Example

Scenario:

- **UAV1** (50kg capacity) and **UAV2** (45kg capacity).
- **Outposts:**
 - Outpost A: 20kg in 30 minutes.
 - Outpost B: 35kg in 60 minutes.
 - Outpost C: 30kg in 40 minutes.

Process:

1. **Priority Calculation:**
A > C > B (based on deadlines).
2. **Path Calculation:**
Shortest paths to each outpost are determined using Dijkstra's algorithm.
3. **First Assignments:**
 - **UAV1:** Delivers 20kg to A and 30kg to B sequentially.
 - **UAV2:** Delivers 30kg to C.
4. **Refuel/Reassign:**
 - **UAV2** returns to base, refuels, and picks up 15kg for A.
5. **Final Deliveries:**
Remaining demands are fulfilled, and logs are updated.

Related Research and Literature Review:

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